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DSCI 550

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Homework 2

1.
 - a. The most specific hypothesis S is the tightest boundary that covers only the positive training examples and nothing more ($20000 \leq \text{Price} \leq 30000$) and ($1600 \leq \text{Engine Size} \leq 2500$)
 - b. The most general hypothesis G is the broadest boundary that covers all examples in the feature space ($12000 \leq \text{Price} \leq 65000$) AND ($990 \leq \text{Engine Size} \leq 3500$)
 - c.
 - i. (a) True Positive
 - ii. (b) True Negative
 - iii. (c) False Negative
 - d. Our hypothesis would need to change since the new point is a negative that falls within the boundary of hypothesis S , so we must either revise our classes further or accept an imperfect classification system based on the training set.
2.
 - a. The model has 50 true positives ($50/100 = \text{recall of } 0.5$)
 - b. The model has 50 false negatives (recall of 0.5)
 - c. The model has 12.5 false positives (0.8 precision, $50/(50+x)$)
 - d. F1 score: $(2 \cdot 0.8 \cdot 0.5) / (0.8 + 0.5) = 0.8 / 1.3 \sim 0.615$
 - e. The reason we prefer high precision and low recall is because it is essential to limit false positives in serious situations (things like criminal cases) and a high precision does so, while sacrificing recall is generally ok.
3. My model would be pass if $\text{midterm} \geq 74$ and $\text{final} \geq 71$ based on the training data. The training accuracy of this model is 92.59% (25/27). We use midterm and final over homework since homework has a much wider range of values for failing students and more overlap with passing students.

4.

$$4. E(A) = \sum_{j=1}^n \frac{s_{1j} + \dots + s_{mj}}{s_m} I(s_{1j}, \dots, s_{mj})$$

$$I(s_1, s_2, \dots, s_m) = - \sum_{i=1}^m \frac{s_i}{s} \log_2 \frac{s_i}{s}$$

$$I(s_1(\text{Yes}), s_2(\text{No})) = \dots$$

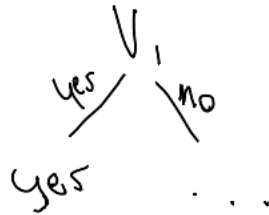
$$I(4, 2) = \left(\left(\frac{4}{6} \log_2 \frac{4}{6} \right) + \left(\frac{2}{6} \log_2 \frac{2}{6} \right) \right)$$

$$I(4, 2) \approx 0.918$$

V1	V2	V3	V4
$I(V1) = \text{Yes} = 0$	$I(V2) = \text{Yes} = 1.0$	$I(V3) = \text{Yes} = 0.811$	$I(V4) = \text{Yes} = 1$
$I(V1) = \text{No} = 0.918$	$I(V2) = \text{No} = 0$	$I(V3) = \text{No} = 1$	$I(V4) = \text{No} = 0$
$E(V1) = \frac{1}{2}(0) + \frac{1}{2}(0.918)$	$E(V2) = \frac{1}{3}(0) + \frac{2}{3}(1)$	$E(V3) = \frac{2}{3}(0.811) + \frac{1}{3}(1)$	$E(V4) = \frac{2}{3}(1) + \frac{1}{3}(0)$
$E(V1) \approx 0.459$	$E(V2) \approx 0.667$	$E(V3) \approx 0.811$	$E(V4) = 0.667$
$G(V1) = 0.459$	$G(V2) = 0.252$	$G(V3) = 0.044$	$G(V4) = 0.252$

cont...

split at root node V_1



Use remaining 3 points (Items 3, 4, 6)

$$G(V_2) = 0.418$$

$$G(V_3) = 0.252$$

$$G(V_4) = 0.418$$

pick one

cont... Final tree

